

ShiftGuard10: Robust Image Classification Challengee

Group 35
EE708 Term project

Abstract—This report presents our approach to the Shift Guard Challenge, focusing on training models entirely from scratch without pretrained weights. We evaluate multiple architectures including Vision Transformers (ViT), deep convolutional neural networks, hybrid CNN-MLP models, and lightweight CNNs. Our study investigates robustness under distribution shifts and analyzes how architectural inductive biases affect generalization. Experimental results demonstrate that transformer-based models outperform convolutional baselines due to their ability to model global dependencies effectively.

I. INTRODUCTION

Robustness to distribution shift is a fundamental challenge in modern machine learning systems. While pretrained models have shown strong performance, they often rely on large-scale datasets and may not generalize well to unseen distributions.

The Shift Guard Challenge emphasizes robustness without reliance on pretrained weights. This constraint highlights the importance of architectural design and training strategies. In this work, we systematically evaluate multiple non-pretrained architectures to understand their behavior under such constraints.

II. DATASET AND PREPROCESSING

We use the CIFAR-10 dataset [1], consisting of 60,000 RGB images of size 32×32 across 10 classes.

To improve generalization, we apply the following preprocessing techniques:

- **Normalization:** Standardizing inputs using dataset mean and variance
- **Random Cropping:** Introducing spatial variation
- **Horizontal Flipping:** Improving invariance
- **Mixup Augmentation:** Combining samples to regularize training [4]

These augmentations are particularly critical when training from scratch, as they compensate for the lack of pretrained feature representations.

III. MODELS

A. Vision Transformer (ViT)

Vision Transformers [2] represent a paradigm shift from convolution-based models by treating images as sequences of patches. Each image is divided into fixed-size patches, which are linearly embedded and processed using transformer encoder blocks.

The core attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Unlike CNNs, ViTs have a global receptive field from the first layer, enabling them to capture long-range dependencies between image regions. Multi-head attention allows the model to learn diverse feature representations across different subspaces.

However, training ViTs from scratch is challenging due to the absence of strong inductive biases. To address this, we employ:

- Learning rate warmup
- Strong data augmentation
- Regularization techniques such as dropout

These strategies stabilize training and improve generalization.

B. Deep CNN

We implement a deep convolutional network inspired by residual architectures [3]. The model consists of stacked convolutional layers with batch normalization and ReLU activations.

Key properties:

- Hierarchical feature extraction
- Local receptive fields
- Strong inductive bias for spatial locality

As depth increases, CNNs gradually capture larger receptive fields. However, this comes at the cost of increased computational complexity and potential vanishing gradient issues.

While deep CNNs perform well on standard datasets, their reliance on local operations limits their ability to model global relationships efficiently.

C. Hybrid CNN-MLP Model

The hybrid CNN-MLP model combines convolutional feature extraction with a fully connected classifier.

Architecture:

- Convolutional layers for feature extraction
- Flattening operation
- Multi-layer perceptron for classification

This design allows efficient feature extraction while maintaining a simple classification pipeline. However, flattening leads to loss of spatial structure, which negatively impacts performance.

Additionally, the MLP introduces a bottleneck, reducing the richness of learned representations.

D. Shallow CNN

To analyze the effect of model capacity, we implement a shallow CNN with fewer layers and parameters.

Characteristics:

- Reduced depth and parameter count
- Faster training
- Limited representational power

While computationally efficient, the shallow CNN struggles to capture complex patterns and exhibits poor robustness under distribution shifts.

IV. TRAINING STRATEGY

All models are trained from scratch using a unified training pipeline to ensure a fair comparison across architectures. This consistency allows us to isolate the effect of model design from training-related variations.

A. Optimization

We use the AdamW optimizer [5], which decouples weight decay from gradient updates, leading to improved generalization compared to standard Adam. This is particularly important in the absence of pretraining, where regularization plays a crucial role in preventing overfitting.

B. Learning Rate Scheduling

A cosine annealing schedule is employed to gradually reduce the learning rate over time. This provides a smooth transition from exploration to convergence and avoids abrupt changes that may destabilize training. The schedule helps models converge to better minima, especially in longer training regimes.

C. Batching and Regularization

All models are trained with a batch size of 128, balancing computational efficiency and gradient stability.

To improve robustness and generalization, we incorporate:

- **Dropout:** Reduces co-adaptation of neurons and prevents overfitting
- **Mixup augmentation** [4]: Generates convex combinations of training samples, encouraging linear behavior between classes and improving robustness under distribution shift

D. Training Stability for Vision Transformers

Training Vision Transformers from scratch is more challenging than CNNs due to the lack of strong inductive biases such as locality and translation invariance.

To address this, we employ a learning rate warmup strategy during the initial epochs. The learning rate is gradually increased from a small value to the target learning rate, preventing unstable updates in the attention layers.

This is particularly important because:

- Attention weights are highly sensitive during early training
- Large gradients can lead to divergence

- Stable initialization improves convergence speed and final performance

Overall, these strategies ensure stable optimization and enable fair comparison between transformer-based and convolutional architectures trained entirely from scratch.

V. RESULTS

Model	Accuracy	F1 Score
Vision Transformer	90.0%	0.88
Deep CNN	79.2%	0.71
Hybrid CNN-MLP	76.6%	0.63
Shallow CNN	72.4%	0.58

TABLE I
PERFORMANCE COMPARISON OF NON-PRETRAINED MODELS

The Vision Transformer achieves the best performance, demonstrating superior generalization capabilities.

VI. DISCUSSION

The results highlight fundamental differences between architectures:

- **Transformers:** Capture global dependencies effectively, leading to improved robustness
- **CNNs:** Strong local feature extraction but limited global context
- **Model Capacity:** Deeper models outperform shallow ones, but require careful regularization

Training from scratch amplifies these differences, making architectural choices more significant.

VII. CONCLUSION

We present a comparative study of non-pretrained models for robust image classification. Vision Transformers outperform convolutional baselines due to their ability to model global interactions.

This work demonstrates that even without pretraining, transformer-based architectures can achieve strong performance when trained with appropriate strategies.

REFERENCES

- [1] A. Krizhevsky, "Learning multiple layers of features from tiny images," 2009.
- [2] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020.
- [3] K. He et al., "Deep residual learning for image recognition," CVPR, 2016.
- [4] H. Zhang et al., "mixup: Beyond empirical risk minimization," ICLR, 2018.
- [5] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," ICLR, 2019.
- [6] A. Vaswani et al., "Attention is all you need," NeurIPS, 2017.